PERIPHERAL ARTERIAL DISEASE / CHRONIC LIMB ISCHEMIA

Overview

- Chronic atherosclerotic progressive arterial occlusion (lower limb mainly). **Age = most important risk factor** (uncommon <40, 25%+ at ≥80). Affects ~10% population; **strong surrogate marker for coronary + carotid disease**.
- **Etiology: atherosclerosis >95%.** Rare (<5%): cystic adventitial disease, popliteal entrapment, **Buerger's (thromboangiitis obliterans)** — non-atherosclerotic segmental inflammatory.

Pathophysiology

- Exercise → O2 demand exceeds supply → anaerobic metabolism + lactate → **intermittent claudication**. Insufficient even at rest → **rest pain = chronic limb-threatening ischaemia (CLTI)**.
- **Tibial pressure <50 mmHg or toe <30 mmHg → severe ischaemia → tissue necrosis.**

Risk factors

- **Smoking (2–4× , strongest), diabetes (2–4×, esp + smoking)**, hypertension, dyslipidaemia (↑cholesterol, ↓HDL), obesity/metabolic syndrome, inflammation (CRP, fibrinogen), low socioeconomic status.

Clinical presentation

- **SFA = most commonly affected**, then popliteal (esp diabetes); below-knee severe.
- **Fontaine / Rutherford classification**:

Fontaine	Rutherford	Clinical
I	0	Asymptomatic
IIa	1	Mild claudication
IIb	2–3	Moderate-severe claudication
III	4	**Ischaemic rest pain**
IV	5–6	Tissue loss (minor/major)

- **Intermittent claudication**: cramp-like pain on walking, relieved by rest. **Calf = SFA disease; thigh/buttock = aorto-iliac.** Occurs when stenosis >70%.
- **CLTI**: rest pain/ulcer/gangrene >2 weeks. Rest pain starts in **toes/forefoot**, worse at night (loss of gravity, ↓CO, warmth steal), **relieved by dangling foot**.
- **Blue toe syndrome ("foot trash")**: microemboli from AAA/popliteal aneurysm/cardiac → cyanotic painful toes.
- **Gangrene**: **dry** (no infection, mummified, demarcated) vs **wet** (infected, polymicrobial, surgical emergency).

Evaluation

- History + exam (palpate all pulses, **Buerger's test/angle**).
- **ABPI** (highest ankle / highest brachial): **<0.7 claudication, <0.3 rest pain/critical**. Exercise ABPI (aorto-iliac; >20 mmHg drop = significant). **TBI** (calcified arteries — DM/CKD; **toe ≥30 mmHg good healing, <30 poor, <20 amputation risk**). TcPO2.

- **Imaging**: **duplex US (first-line)** (>70% = monophasic); CT/MR angiography; **catheter angiography = gold standard** (when intervention planned).

Management

- **Natural history of claudication: 1/3 improve, 1/3 stable, 1/3 worsen.**
- **Conservative**: **smoking cessation**, BP <140/90, statins, glycaemic control, **supervised exercise (30–45 min, 3–4×/wk, ≥12 wks)**, weight loss.
- **Best medical therapy**: **antiplatelet (aspirin = COX-1/thromboxane; clopidogrel = P2Y12)**, **statin (all patients)**, vasoactive (**naftidrofuryl** = 5-HT2 antagonist; **cilostazol** = PDE-III inhibitor), prostanoids (iloprost) for severe.
- **Revascularization**:
- **Indications**: absolute = CLTI (rest pain/tissue loss/gangrene); relative = disabling claudication <50m, lifestyle-limiting, failed conservative.
- **Endovascular**: angioplasty, stent, atherectomy (less invasive, faster recovery).
- **Open (gold standard for complex)**: **endarterectomy + patch** (femoral, profundaplasty); **bypass** (aorto-bifemoral, fem-fem crossover, fem-popliteal, distal; **vein conduit preferred**).
- **Hybrid** for multilevel.
- **Amputation** (non-reconstructable, sepsis, frail); **lumbar sympathectomy** (rare, ineffective in DM); stem cell therapy (experimental).

High-yield one-liners

- **Atherosclerosis >95%; smoking + diabetes strongest risk; SFA commonest site.**
- **Fontaine/Rutherford; claudication at >70% stenosis; CLTI = rest pain/ulcer/gangrene >2wks.**
- **ABPI <0.7 claudication, <0.3 critical; TBI for calcified arteries (DM/CKD).**
- **Rest pain relieved by dangling foot; dry vs wet gangrene (wet = emergency).**
- **Conservative 1/3 improve; cilostazol (PDE-III), naftidrofuryl (5-HT2); revascularize for CLTI.**

CHRONIC LIMB ISCHEMIA (PAD) — Revision Table

Bold = high-yield.

OVERVIEW & CLASSIFICATION

Topic	Key points
Etiology	**Atherosclerosis >95%**; rare = Buerger's, popliteal entrapment
Risk factors	**Smoking + diabetes (strongest, 2–4×)**, HTN, dyslipidaemia, age (most important)
Pathophysiology	**Tibial <50 mmHg / toe <30 mmHg = severe ischaemia**; SFA commonest site
Fontaine/Rutherford	I/0 asymptomatic; IIa/1 mild, IIb/2-3 mod-severe claudication; **III/4 rest pain; IV/5-6 tissue loss**
Claudication	**Calf = SFA; thigh/buttock = aorto-iliac**; at >70% stenosis
CLTI	Rest pain/ulcer/gangrene **>2 weeks**; rest pain in toes, worse at night, **relieved by dangling foot**
Gangrene	**Dry (no infection) vs wet (infected = emergency)**; blue toe = microemboli

EVALUATION & MANAGEMENT

Topic	Key points
ABPI	**<0.7 claudication, <0.3 critical**; exercise ABPI (aorto-iliac, >20 mmHg drop)
TBI	Calcified arteries (DM/CKD); **toe ≥30 good healing, <30 poor, <20 amputation**
Imaging	**Duplex US first-line**; **catheter angiography = gold standard**
Natural history	**1/3 improve, 1/3 stable, 1/3 worsen**
Conservative	**Smoking cessation, statins, supervised exercise** (30–45min, 3–4×/wk)
BMT	**Antiplatelet (aspirin/clopidogrel) + statin**; cilostazol (PDE-III), naftidrofuryl (5-HT2)
Revascularization	**CLTI = absolute indication**; endovascular (angioplasty/stent) vs open (endarterectomy/bypass — **vein preferred**); hybrid
Other	Amputation (non-reconstructable/sepsis); lumbar sympathectomy (rare)